

Keeping Bias Out of Lending AI

Deterministic audits, LLM interpretation, and what the standard toolchain leaves to you

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1 THE QUESTION

Models now decide consumer installment credit and residential mortgage origination. Both are regulated — ECOA and Regulation B govern both, the Fair Housing Act adds to mortgage. **What must a team actually do to keep bias out of a model in these sectors, how much of that does the standard fairness toolchain do for them, and can a language model safely do the rest?**

2 WHY IT MATTERS

Four decisions determine whether a lending audit is *legally* meaningful: which outcome counts as favorable, who counts as protected, how much data supports each subgroup estimate, and against whom the comparison runs. Get any one wrong and the audit still runs, still emits a number, and still reads clean — the direction nobody re-examines. A false alarm costs a reviewer an afternoon; a false clearance closes the question and the applicant never learns why.

3 WHAT WE DID

A complete audit of both sectors on real data: 1,000 consumer credit and 10,978 Georgia mortgage applications, every applicant scored out-of-fold by a model that never saw them. The architecture separates two jobs.

Deterministic layer. Python owns every number: AIF360 computes the metrics, a fixed rule assigns severity. This is the ground truth.

LLM layer. The model owns only interpretation — who is affected, what mechanism, what remedy. It never recomputes a metric, and any number in its output absent from the given context is recorded as fabrication. Calculator, not oracle.

Benchmark. Two tracks, kept separate. *Track Q*: four prompt strategies per attribute, severity scored against the deterministic label. *Track H*: narrative audits judged only on a checklist and rubric, never against the numbers. 98 calls, \$0.89, cumulative cost cap, offline replay without an API key.

4 HOW IT IS RELEVANT

The toolchain leaves the legal questions open. Verified directly against AIF360 0.6 and Fairlearn 0.13: AIF360 defaults `favorable_label=1.0` so on a target named `loan_default` it treats defaulting as the good outcome; Fairlearn exposes no favorable-label parameter at all. Neither library has *any* concept of minimum subgroup size — no floor, no warning, no guidance. Neither binds a protected class to a statute.

Intersectional cells stay unidentified even at full population. Six of twelve HMDA race×sex cells have no determinate severity band: American Indian × Male is 0.6734 with a 95% interval of [0.45, 0.90] at $n=31$ — consistent with a critical violation and with passing. Growing the audit fivefold did not resolve them.

The defensible disparities are the large cells: Black women and men at 0.863/0.864, intervals wholly below parity, which a point-estimate ranking demotes to sixth and seventh.

The statute matters more than the metric. Auditing age at 40+ (ADEA, employment law) returns 0.9791 — clean. At 62+, the cut ECOA specifies, the same model returns **0.8897**: the largest age disparity present.

An LLM is a usable interpreter, not a usable measurer. Aggregate agreement of 75% decomposes to 33%/92%/100% and tracks task difficulty rather than capability. Grounding the prompt in retrieved literature cut fabricated citations $16 \rightarrow 2$, a larger effect than any prompt strategy. Prompt sensitivity is generation-specific: spread across four strategies is 14.7 points on gpt-4o and 0.0 on gpt-5-mini.

5 HOW IT APPLIES

1. State the favorable outcome explicitly; never accept the default. 2. Audit the book, not a test split — in mortgage, HMDA already requires filing it. 3. Put the statute citation beside the threshold in code: ECOA 62+, not ADEA 40+; note that §1002.6(b)(2) permits *favouring* elderly applicants, so a ratio above 1 is lawful and a symmetric metric cannot express that. 4. Publish what was suppressed, with n and the unreported ratio. 5. Record the reference group beside every ratio. 6. Let the model interpret, never compute, and score it against numbers it cannot move.

6 THE PROOF

Table 1: Every claim regenerates offline from the released repository.

Claim	Where to check it
AIF360 default inverts a risk target	Minimal repro: default 1.2072 vs. correct 0.9862; RESULTS.md
6/12 cells band-indeterminate	scripts/interval_estimates.py
ECOA 0.8897 vs. banded 0.9791	hmda/fairness/fairness_summary.json
Agreement 33/92/100; Grounding 16→2	consolidated/track_q_analysis.json
Cross-model spread 14.7 vs. 0.0	llm_benchmark/gpt-5-mini/
\$0.8933 over 98 calls	llm_benchmark/spend_ledger.json

60 tests, CI green; the deterministic layer regenerates with no API key.